Digital Communication IC Design

Homework #3

(Due on 2026/11/04)

- Please upload your codes and reports (with simulation results and required explanations) to the NTUCool.
- Please disclose how you use AI tools and its version.

Total points: 130 points

1. Consider the four-path channel power delay profile below. Use linear power, not dB values, when calculating channel statistics.

Tap i	Power $P_{i,\mathrm{dB}}$	Delay τ_i
0	0 dB	0 ns
1	-2 dB	16 ns
2	-6 dB	92 ns
3	-9 dB	200 ns

(a) Convert every path power to linear scale, normalize the four powers so that they sum to one, and print both vectors, $\mathbf{p} = [p_1\ p_2\ p_3\ p_4]$ **and** $\tilde{\mathbf{p}} = [\tilde{p}_1\ \tilde{p}_2\ \tilde{p}_3\ p_4]$ with at least six significant digits. (10%)

$$p_i = 10^{P_{i,\mathrm{dB}}/10}, \tilde{p}_i = \frac{p_i}{\sum_r p_r}$$

(b) Calculate the mean excess delay, RMS delay spread, and maximum excess delay. Show the contribution of each path to the first and second delay moments. (10%)

$$\overline{\tau} = \sum_i \tilde{p}_i \tau_i\,, \tau_{\mathrm{RMS}} = \sqrt{\sum_i \tilde{p}_i (\tau_i - \overline{\tau})^2}$$

(c) Using the definition below, calculate the coherence bandwidth. Compare it with signal bandwidths of 2.5 MHz and 250 MHz and classify each channel as approximately flat fading or frequency selective. Explain the criterion. **(5%)**

$$B_c = \frac{1}{5\tau_{\mathrm{RMS}}}$$

(d) At sampling rate Fs = 250 MHz, convert every path delay to an integer sample delay. Determine the discrete channel length and the minimum cyclic-prefix length that prevents inter-symbol interference for this channel. **(5%)**

2. Let m be the last two digits of the student ID and use m as the random seed. Generate independent γ_i distributed as CN(0,1) (complex normal distribution), with real and imaginary components each having variance 1/2. Use the powers and delays from Question 1.

(a) Generate the four complex path gains and normalize them using the equation below. Print γ_i, g_i, h_i, their magnitudes and phases, and the final sum of squared magnitudes. **(15%)**

$$g_i = \sqrt{p_i}\gamma_i, h_i = \frac{g_i}{\sqrt{\sum_r |g_r|^2}}$$

(b) The continuous-time impulse response is given as follows. Derive its continuous-frequency response. Draw the frequency domain response including magnitude and phase versus frequency. Note that the frequency range should cover -250MHz to 250MHz. (15%**)**

$$h(t) = \sum_{i=0}^{3} h_i \delta(t - \tau_i)$$

(c) How do you verify your figure in Q2(b)? Do you observe any periodicity? Please state the reasons for the periodicity or non-periodicity of the figure. (10%)

3. Use **Fs = 250 MH** and the time-invariant discrete channel from Question 2 to design an OFDM system.

(a) Determine the minimum guard interval length to prevent ISI, and select an appropriate N_g value that is a power of 2 (10%)

(b) If the guard interval ratio should be smaller than 1/4, what is the proper setting of FFT size N in an OFDM system. Select the minimum value to satisfy the constraint. (5%)

(c) Now calculate the subcarrier frequency f_k in the frequency domain for $k = -N/2 \sim N/2 - 1$. Draw the frequency domain response at each subcarrier of the channel described in Q2. Note that the frequency response should include both phase and magnitude. Please use stem plots. (10%)

4. Now, let transmit the OFDM signal through the channel. Use the parameters calculated in Q3.

(a) Please randomly generate frequency domain data from QPSK constellation (black constellation points in Homework 1). Draw the real part and imaginary part of frequency domain data $X[k]$ versus subcarrier index. Then transform it to time domain and add the cyclic prefix with length specified in Q3, which is indicated by $x[n]$. (10%)

(b) Use your own channel generated in Q2. Assume the discrete-time channel impulse response is described by

$$h[n] = \sum_{i=0}^{3} h_i \delta[n - n_i], n_i = \tau_i F_s.$$

Then generate the received signal $\mathrm{y}[n] = \mathrm{h}[\mathrm{n}] \otimes \mathrm{x}[\mathrm{n}]$, where $\otimes$ denotes linear convolution. Then, draw the real-part and imaginary-part of your time-domain received signal first. Remove CP, transform it back to frequency domain by DFT to obtain $Y[k]$. Draw the magnitude of your frequency domain signal versus subcarrier index. (10%)

(c) Please calculate

$$H[k] = \frac{Y[k]}{X[k]}$$

Draw the magnitude of the frequency domain response $H[k]$ (10%). Observe this figure and the figure in Q3(c). Are they the same or not the same? Please explain the reason. (5%)